

Full Stack Web Development Internship Report

Project: StayScape (Accommodation Booking Platform)
Vaibhav Patel • B.Tech CSE (AI/ML) • IILM University

Candidate & Program Overview

Student Details

- ▶ Name: Vaibhav Patel
- ▶ Degree: B.Tech CSE (AI & ML)
- ▶ Roll Number: 25scs1003003209
- ▶ Year/Sem: 2nd Year, 3rd Semester
- ▶ Institution: IILM University, Greater Noida

Internship Profile

- ▶ Organization: Apna College
- ▶ Program: Delta (Web Development)
- ▶ Duration: 14 Weeks
- ▶ Timeline: Jun 03, 2026 - Sep 10, 2026
- ▶ Status: Successfully Completed

Organization Profile

Apna College is a premier educational technology organization focused on equipping engineering students with industry-relevant technical skills.

- ▶ ***Rigorous Curriculum:** Known for practical, hands-on teaching methodology bridging theoretical computer science with industry software development.*
- ▶ ***Delta Program Focus:** Deep-dive into the modern JavaScript stack, training developers to create scalable, production-ready applications.*
- ▶ ***Industry Approach:** Students architect, code, and debug real-world applications (like StayScape) from the ground up.*



Problem & Scope

The Challenge

The travel industry relies heavily on digital platforms. However, developing secure, scalable architectures capable of managing dynamic user-generated content, media processing, and complex relational data in a NoSQL environment is inherently difficult.

The StayScape Solution

The project scope was to engineer *StayScape*, a robust platform mirroring industry-leading applications. It features secure authentication, dynamic property listing CRUD operations, an interactive review system, and seamless cloud-based media integration.

Core Project Features



Secure Authentication

Session-based authentication with role-based access control. Ensures only property owners can modify or delete specific listings securely.



Dynamic CRUD

Seamless creation, reading, updating, and deletion of property listings. Includes rigorous server-side schema data validation.



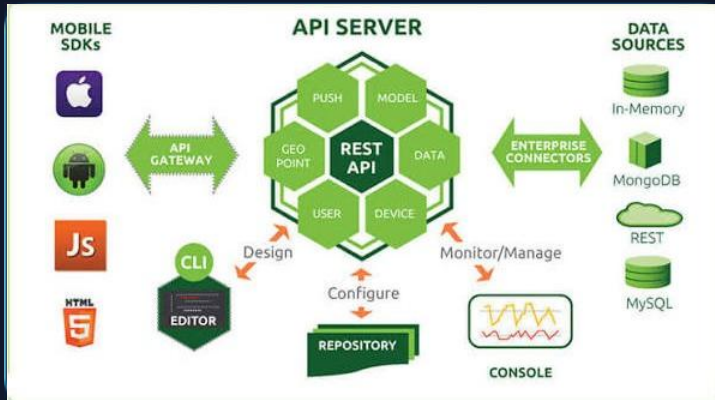
Cloud Media Integration

Direct image uploading capabilities utilizing cloud storage APIs, bypassing local limitations for fast asset delivery.

Tools & Techniques Used

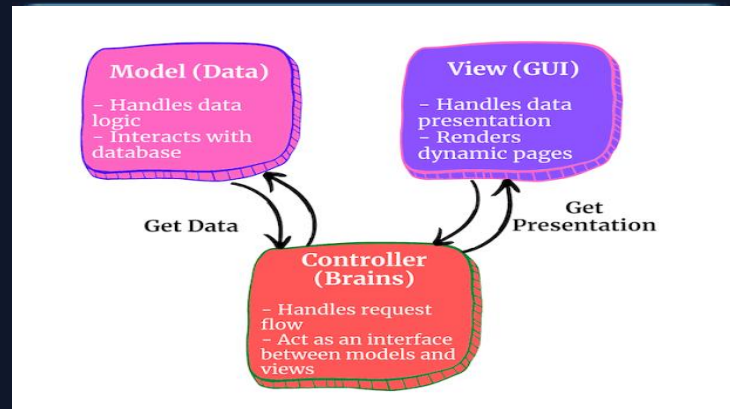
Development Layer	Technologies & Frameworks	Primary Purpose
Frontend (UI/UX)	HTML5, CSS3, JS, EJS, Bootstrap 5	Responsive, mobile-first design and dynamic templating
Backend (Server)	Node.js, Express.js	Robust server creation and RESTful API routing
Database (Data)	MongoDB, Mongoose ODM	NoSQL data management and strict schema validation
Security & Storage	Passport.js, Joi, Multer, Cloudinary	Authentication, data validation, and remote asset storage

Project Architecture (MVC)



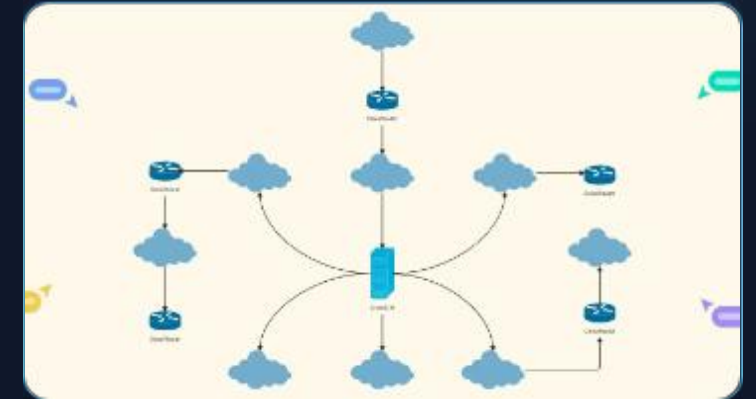
1. Models

Defines DB structure. Contains `listing.js` (Properties), `review.js` (Feedback), and `user.js`.



2. Views

Handles presentation. Uses EJS templates (`index`, `show`) to render dynamic backend data.



3. Controllers

Operational logic. Intercepts HTTP requests, interacts with Models, and serves the Views.

Internship Project Timeline

Jun 26 - Jul 15

Phase 2: Backend

Node.js & Express setup.
Routing HTTP requests and
RESTful API design.

Sep 01 - Sep 10

Phase 4: Auth & Deploy

Passport.js security, Cloudinary
uploads, bug fixing, and
finalization.

Jun 03 - Jun 25

Phase 1: Foundation

Core web logic; understanding
DOM manipulation and
asynchronous JavaScript.

Jul 16 - Aug 31

Phase 3: Database & Core

MongoDB integration, MVC
setup, and developing full CRUD
logic for listings.

Skills & Learning Outcomes

 **Full-Stack Proficiency** Successfully bridged responsive frontend user interfaces with complex backend logic, achieving a holistic understanding of the web lifecycle.

 **Advanced Database Design** Acquired the ability to design relational-style data models within a NoSQL environment (embedding ObjectIDs of reviews within property schemas).

 **Middleware Implementation** Mastered custom middleware for request logging, authentication checks, validation routing, and robust error interception.



Official Certification

The rigorous training and project development culminated in the successful awarding of the "Delta (Full Stack Web Development)" certificate by Apna College.

This certification serves as a formal validation of the technical competencies acquired during the program and the successful completion of the StayScape capstone project.



Conclusion

This project-based training provided an invaluable simulation of a professional software development environment. By architecting, coding, and debugging a complex, data-driven web application from scratch, a profound understanding of the modern web development lifecycle was achieved. The technical skills and problem-solving methodologies cultivated during this period—spanning frontend design, backend logic, and database management—have established a robust foundation for future academic endeavors and a professional career in software engineering.